Exp No: 12 Experiment to detect outliers in a given data set.

AIM:

To conduct an experiment to detect outliers in a given data set.

PROGRAM:

```
import numpy as np
array = np.random.randint(1, 100, 16)
array
```

```
array([45, 51, 89, 18, 64, 14, 93, 27, 23, 52, 93, 50, 42,
2, 85,  6])
```

```
array.mean()
```

```
np.float64(47.125)
```

```
np.percentile(array, 25)
```

```
np.float64(21.75)
```

```
np.percentile(array, 50)
```

```
np.float64(47.5)
```

```
np.percentile(array, 75)
```

```
np.float64(69.25)
```

```
np.percentile(array, 100)
```

```
np.float64(93.0)
```

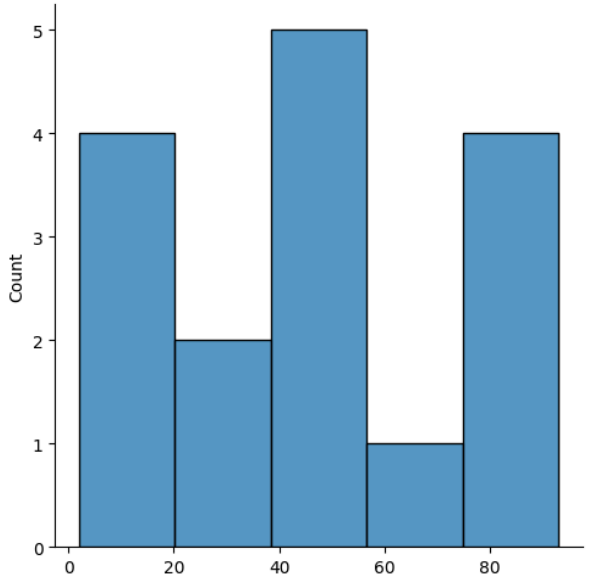
```
def outDetection(array):
   sorted(array)
```

```
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur
lr, ur = outDetection(array)
lr, ur
```

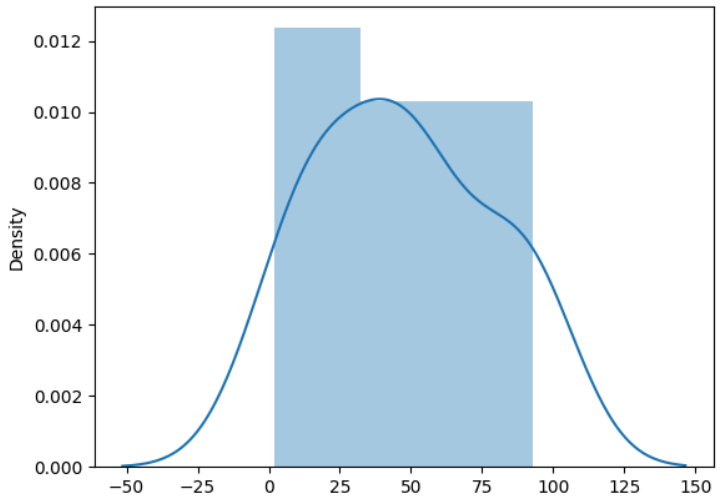
```
(np.float64(-49.5), np.float64(140.5))
```

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

```
<seaborn.axisgrid.FacetGrid at 0x7cdd594dcad0>
```



sns.distplot(array)



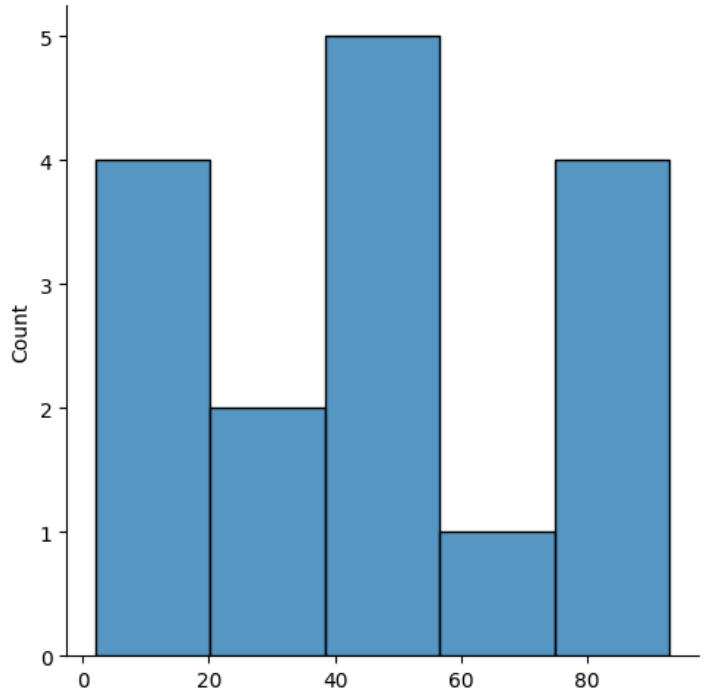
new_array = array[(array > lr) & (array < ur)]

new_array

```
array([45, 51, 89, 18, 64, 14, 93, 27, 23, 52, 93, 50, 42,
2, 85,  6])
```

sns.displot(new_array)

<seaborn.axisgrid.FacetGrid at 0x7cdd5930c550>



lr1, ur1 = outDetection(new_array)

lr1, ur1

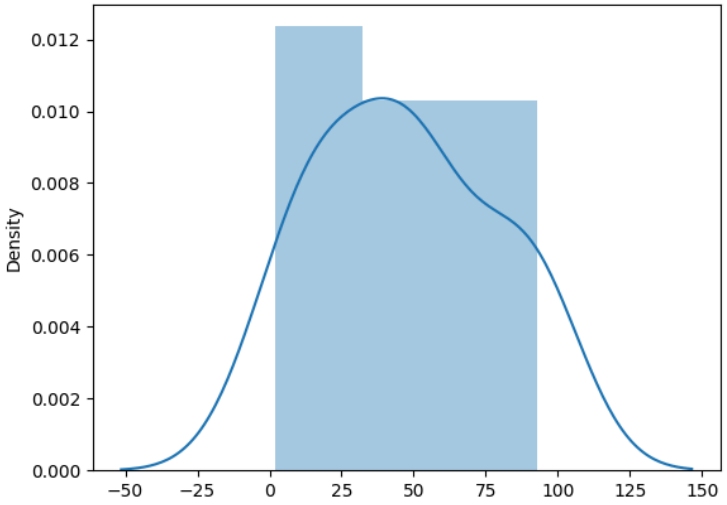
```
(np.float64(-49.5), np.float64(140.5))
```

final_array = new_array[(new_array > lr1) & (new_array < ur1)]

final_array

```
array([45, 51, 89, 18, 64, 14, 93, 27, 23, 52, 93, 50, 42,
2, 85,  6])
```

sns.distplot(final_array)



RESULT:

Thus an experiment to detect outliers in a given data set was conducted.